

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

PLANT CHECKLIST

Controls Commissioning Checklist: FAT and SAT for Aggregate, MRF, and Organics/AD Plants

Field notes from Jonathan Gilmour. This is the checklist I actually run off of when I'm proving out a control system, either on the panel-shop floor (Factory Acceptance Test, FAT) or after it's bolted down and wired in at the plant (Site Acceptance Test, SAT). It's built to work across three plant types I see most: aggregate (crushing, screening, conveying), material recovery facilities (MRF) with their sort lines and balers, and organics / anaerobic digestion (AD) with biogas handling. Print it, mark it up, sign it. A commissioning record that lives only in your head is worth nothing at the next turnover.

Two rules before anything else. First: you prove an interlock by making it trip, not by talking yourself past it. If a protective interlock is in your way during commissioning, you fix the logic or the setpoint, you don't jumper it out to "get through the day." Second: every force, every temporary jumper, every defeated guard gets written on the force log and removed before handover. A single forgotten force has hurt more people than any software bug.

Header note: gas safety and LOTO (read before the hands-on sections)

Organics / AD plants: digesters, gas holders, sumps, pump vaults, and wet wells are permit-required confined spaces. Do not enter or open one on the strength of "it smells fine." Hydrogen sulfide (H₂S) kills fast and deadens your sense of smell around ~100 ppm, which is also roughly the IDLH (Immediately Dangerous to Life or Health). Methane (CH₄) is explosive from about 5% by volume (Lower Explosive Limit, LEL) up to about 15% (Upper Explosive Limit, UEL). Atmospheric-test before entry and wear a bump-tested personal 4-gas monitor (H₂S, LEL, O₂, CO). Biogas areas are typically Class I Div 1 or Div 2 (or Zone 0/1/2) under NEC 500/505 — your test gear, lights, and any temporary tooling in those areas must be rated for it. That includes the meter you're using to megger or the laptop you're not supposed to have in there.

All plant types: any hands-on work on rotating equipment (conveyors, crushers, screens, augers, mixers, balers, grinders) requires a verified zero-energy state — Lockout/Tagout (LOTO) applied and tested-for-dead — before you put a hand on it. Commissioning is exactly when a drive spins that

nobody expected, because the logic is half-written. During bump tests and rotation checks, keep bodies out of pinch points and nip points and use a spotter with a clear line to the disconnect.

Where a step below involves opening an energized enclosure or running an energized test, the hazard note sits above the step. Treat it as part of the step, not a footnote.

1. Pre-power (before you put a single volt on it)

- ☐ Drawings on hand and current: schematics, panel layout, I/O list, network/architecture, cause-and-effect (C&E) matrix. Note the revision on each. If FAT drawings and site conditions already disagree, flag it now.
- ☐ Terminations: every field and internal termination torqued to spec and marked (a paint pen stripe or torque-check sticker). Tug-test a representative sample; a loose lug that passes point-to-point today is a thermal fault next summer.
- ☐ Wire numbering matches the schematic end to end. Ferrules present and legible.
- ☐ Segregation: 120/480 VAC power kept away from 24 VDC control and from analog/signal wiring. Analog and communication cables not run in the same bundle as VFD (Variable Frequency Drive) output cables.
- ☐ Shielding: instrument cable shields landed at ONE end only (per the drawing — usually the panel/control side). Both-ends-grounded shields make a ground loop that shows up later as noise on your 4-20 mA loops.
- ☐ Grounding: panel ground bar bonded to building/plant ground. Equipment grounding conductor continuity checked. VFD, drive, and motor grounds per the drive manual — biogas and dusty-aggregate environments both punish a sloppy ground.
- ☐ Panel breathing and environment: correct NEMA/UL enclosure type for the location; door gaskets intact; cable entries sealed. Thermostat/heater and fan/filter or A/C present where the ambient calls for it. Confirm the panel isn't sited where crusher dust, wash-down, or corrosive digester gas atmosphere will eat it.
- ☐ For any panel in or near a classified (biogas) area: confirm rating and purge/pressurization scheme if used. If it's a general-purpose panel that ended up in a Div 2 zone, stop and get that resolved.
- ☐ Fuses/breakers match the drawing values. Spare fuses stocked.
- ☐ Insulation resistance (megger) on power feeders/motor circuits as required. Disconnect the VFD/soft starter, PLC I/O, power-factor correction capacitors, and surge/EMI suppression devices first — the megger test voltage (often 500–1000 VDC) will punch through and destroy drive and controller electronics still on the circuit. Do NOT megger inside a classified (biogas) area unless it has been de-energized and gas-tested clear or your instrument is rated for the area. **Hazard: this is an energized-source test on isolated conductors; verify isolation before you connect,**

discharge the conductors to ground afterward (the cable holds a charge), and keep others clear.

2. Power-up and I/O checkout

Hazard: this section energizes the panel. Only qualified personnel inside an open, energized enclosure; use proper PPE and keep one hand out where practical during live checks.

- ☐ Bring up power in stages: control transformer/24 VDC supplies first, verify voltages at the supply and at the far end of a long run (voltage drop on 24 VDC over a long conveyor gallery is real).
- ☐ PLC/controller boots clean, no fault LEDs, correct program and revision loaded. Note firmware versions.
- ☐ Confirm the correct project is in the controller and it matches your master copy. A stale program is the classic FAT-to-SAT trap.

Point-to-point I/O checkout (the heart of it):

- ☐ Every discrete input: actuate the field device (press the e-stop, trip the pull-cord, block the photo-eye, lift the level float) and confirm the right bit lights at the right address in the program. No "checked by inference."
- ☐ Every discrete output: command it from the controller and confirm the right device moves/energizes. Verify it's the RIGHT device — swapped conveyor outputs are common and dangerous.
- ☐ Every analog input scaled and verified at multiple points (see calibration below), not just at zero.
- ☐ Every analog output driven to known values and confirmed at the device (e.g., VFD speed reference, damper position).

Force discipline:

- ☐ Start a written force log the moment you place the first force. Address, reason, who, timestamp.
- ☐ Forces are for checkout only. Never leave a force in to "hold" a nuisance input.
- ☐ Before you leave power-up: pull up the controller's force table and confirm it's empty (or every remaining force is on the log with a reason and a removal plan). Physically confirm forces disabled.

Analog scaling and NAMUR / live-zero checks:

- ☐ Confirm 4-20 mA loops use live zero: 4 mA = low end, 20 mA = high end. A signal reading 0 mA is a broken loop, not a real zero.
- ☐ Verify break/fault detection: for transmitters and inputs supporting NAMUR NE43, confirm the logic treats <3.6 mA and >21 mA as a fault (sensor/loop failure), not as a valid process value. Simulate an out-of-range signal and watch the alarm fire — a digester level "reading zero" because the loop is cut must NOT be trusted as "tank empty, keep pumping."

- ☐ RTD/thermocouple inputs: correct type selected, open-circuit detection behaves.
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3. Loop and instrument calibration

- ☐ Each instrument calibrated or field-verified against a known reference at low / mid / high (e.g., 0 / 50 / 100%). Record as-found and as-left values. Note serial numbers and cal-due dates.
 - ☐ Engineering units in the HMI/PLC match the instrument range exactly. A transmitter set 0-15 psi displayed as 0-10 will read wrong all day and nobody catches it until a decision goes bad.
 - ☐ Level, pressure, flow, temperature, gas-analysis (for AD: CH₄ / H₂S / O₂ analyzers) each verified against spec — **check against your machine's and instrument's own data sheet; don't trust a "typical" number from memory.**
 - ☐ Belt scales / weigh systems (aggregate, MRF): zero and span per manufacturer; note the calibration weight or chain used.
 - ☐ Loop check ties the whole path: field value -> transmitter -> PLC -> HMI display -> historian tag all agree.
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4. Drives, motors, rotation, and overloads

Hazard: rotation checks spin equipment. Confirm the area is clear, guards are on where they can be, and a spotter is at the disconnect. LOTO comes off only for the controlled bump, and goes back on for any hands-on adjustment.

- ☐ Before the first bump, prove the stop: confirm the local disconnect AND the e-stop that covers this drive actually drop it. Don't spin a motor you can't reliably stop from where you're standing. (Full e-stop and pull-cord proving is Section 5; this is the minimum for the drive in front of you.)
- ☐ Motor nameplate data entered into each VFD/soft starter: FLA (Full Load Amps), voltage, frequency, RPM, service factor. Overload set to the motor's rated current per spec, not left at default.
- ☐ Bump test each drive for correct rotation BEFORE loading it. Conveyor belts, augers, and pumps run backward do real damage — a reversed digester feed pump or a backward screw conveyor plugs or breaks things fast. Confirm direction against the arrow and the process, then correct by parameter or wiring (and re-verify).
- ☐ Verify overload/trip actually trips: confirm the overload and the drive fault path stop the motor and annunciate. Don't assume the electronic overload works because it's new.
- ☐ Acceleration/deceleration ramps sane for the load; check for nuisance overcurrent trips on start of high-inertia loads (large screens, flywheeled crushers).
- ☐ Local/Remote and HOA (Hand-Off-Auto) behavior correct at the starter and in the logic. Confirm Hand mode still respects safety interlocks (e-stops, guards) — Hand is not a bypass.

- ☐ Zero-speed / motion sensing on conveyors and elevators reads correctly (belt-slip and stall detection depends on it).

5. Interlock and permissive proving (do not skip, do not bypass)

This is where commissioning earns its keep. **Prove each protective interlock by actually causing the condition and watching it stop what it's supposed to stop.** A permissive that was never tripped is an assumption, not a protection.

- ☐ E-stops: trip EACH e-stop and pull-cord individually and confirm it drops the correct zone/equipment, and that reset requires deliberate action (not auto-restart). Walk the whole pull-cord run — MRF sort lines and long aggregate conveyors have many.
- ☐ Guard/gate/door interlocks: open each and confirm the associated machine cannot run.
- ☐ Zone/load-flow interlocks: confirm a downstream stop propagates upstream (see sequencing). Confirm you cannot start a conveyor discharging onto a stopped conveyor.
- ☐ Process-protection interlocks: high level, high pressure, high temperature, high vibration, plugged-chute, jam/stall — each one forced to condition and confirmed to stop/prevent the right thing.
- ☐ AD-specific safety interlocks (**these are safety functions — retune only within the equipment's own limit, one change at a time, with a record; never defeat them to keep gas flowing**): high/low gas pressure, flame arrestor / thermal safety on the flare, gas-detection trips, overpressure relief, foam-over / high-level on the digester, low-flow or high-temp on gas blowers.
- ☐ Nuisance vs. genuine: if an interlock trips too easily, the fix is to retune the setpoint or filter/debounce it WITHIN the equipment's rated limit, documented, one change at a time — not to widen it past the limit and not to jumper it. Note the as-found and as-left setpoint.
- ☐ Record for each: interlock ID, how you tripped it, what it did, pass/fail, and the setpoint. This table is the most valuable page of the whole record.

6. Sequencing

- ☐ Start-from-discharge: confirm an auto start sequences from the discharge end back toward the infeed, with the correct interlock/permissive and time delay between each unit. Nothing feeds onto a belt that isn't running yet.
- ☐ Stop-from-infeed (normal stop): confirm a normal auto stop shuts the infeed first and lets the line run out material before stopping downstream. A "clean stop" that leaves belts loaded is a Monday-morning restart nightmare.
- ☐ Emergency stop vs. normal stop behave differently and correctly (e-stop drops everything now; normal stop sequences out).

- ☐ Jam / stall / plug handling: simulate a stall (zero-speed, motor overload, plugged-chute switch, high-level in a surge bin) and confirm the sequence stops the upstream feed to prevent pile-up and does not try to auto-restart into a jam.
 - ☐ Restart after a trip requires the operator to clear/acknowledge — no surprise auto-restart of a conveyor with someone clearing the jam. This overlaps LOTO: the logic must never be the only thing standing between a tech and a belt.
 - ☐ Timers and permissives match the C&E matrix. Note any field-adjusted delay.
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7. Alarm test

- ☐ Every alarm annunciates: force each alarm condition and confirm it shows on the HMI, logs, and (where used) sounds/lights/notifies.
 - ☐ Priority and consequence are sane: a safety/gas alarm outranks a "filter dirty" advisory. Confirm the priority in the system matches the real-world consequence. Flood of low-value alarms during an upset is how the important one gets missed.
 - ☐ Alarm text is clear and actionable — a tech at 2am should know what tripped and where, not decode a tag number.
 - ☐ Acknowledge, silence, and reset all work as intended. Confirm an alarm re-annunciates if the condition returns after acknowledge.
 - ☐ Gas-detection and safety alarms (AD): confirm they route to where someone will actually see/hear them, including any auto-dial or plant-wide horn.
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8. Failure-mode / power-loss test (prove it fails safe, not fail-dangerous)

The whole point: when something breaks, the plant lands in a safe state on its own.

- ☐ Power blip / brownout: cut and restore control power. Confirm equipment does NOT auto-restart into motion without operator action, controller reboots to a safe state, and nothing latches "on" through the dip.
- ☐ Comms loss: pull a network cable / drop a remote I/O rack / kill a drive network and confirm the affected outputs go to their defined safe state (usually stop), an alarm fires, and recovery is orderly. Verify watchdog/hold-last-state behavior is what the design intended — hold-last-state on a running conveyor during comms loss can be fail-dangerous; confirm it's actually the safe choice for that output.
- ☐ Sensor fail: for critical measurements (digester level/pressure, gas analysis, chute/level switches), simulate a failed sensor (out-of-range per NAMUR, or physically disconnected) and confirm the logic treats it as a fault and moves to safe, not that it keeps running on a bad number.

- ☐ Air/hydraulic loss (where applicable): confirm valves/actuators go to their fail-safe position (fail-closed / fail-open per the design and the hazard).
 - ☐ UPS/backup (if fitted): confirm the controller and critical I/O ride through as intended and that the plant sheds load gracefully.
 - ☐ Record what actually happened for each test versus what was supposed to happen. "We assume it fails safe" is not a test result.
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9. Historian and records capture

- ☐ Critical tags are actually historizing: pull each key value up in the historian/trend and confirm it's logging at a useful rate.
 - ☐ Timestamps correct and synchronized (PLC, HMI, historian, SCADA all on the same clock/time source). Skewed clocks make an incident impossible to reconstruct.
 - ☐ Alarm/event journal is capturing acknowledge and reset events, with operator ID where required.
 - ☐ Retention set to something that outlives a typical investigation window. Confirm backups run.
 - ☐ For AD/regulated streams: confirm the tags a permit or reporting requirement depends on are logging and retrievable.
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10. Documentation, red-lines, and handover

- ☐ Force log: empty and signed off. Confirm one more time no forces remain.
- ☐ Red-line drawings: every field change marked on the schematics/I/O list during FAT and SAT, initialed and dated. Get these back to the drawing owner for as-builts — don't let them die in a binder.
- ☐ Setpoint record: every interlock/alarm setpoint as-left, especially any nuisance setpoint you retuned, with the equipment limit noted alongside so the next person knows the ceiling.
- ☐ Calibration records: as-found/as-left for each instrument, cal-due dates.
- ☐ Interlock/permissive proving table completed and signed.
- ☐ Software: final program, HMI application, drive parameter files backed up and version-labeled, copies handed to the plant AND kept off-site. A commissioning with no saved backup is one lightning strike from starting over.
- ☐ Punch list: open items captured with owner and target date; distinguish "must fix before run" from "fix at next window."
- ☐ Operator/maintenance walk-through done: e-stops, normal vs. emergency stop, alarm response, and the safe-state behaviors from Section 8 shown to the people who'll run it.

- ☐ Sign-off: FAT and SAT each signed and dated by the responsible parties. Note what was tested at FAT and what was deferred to SAT (some things — real material, real gas, full rotation under load — only prove out on site).
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A short note on independent commissioning

Most of the failures I get called back for aren't exotic — they're the interlock nobody tripped, the force nobody removed, the reversed pump nobody bumped, the sensor-fail case nobody simulated. This checklist is vendor-neutral on purpose. If a system is being handed between an equipment supplier, a panel builder, and a plant crew, the seams between them are where things get assumed rather than proven. An independent second look helps most exactly on those hand-off faults, where each party reasonably thought the other one checked it.

Adapt this to your plant. It's a working document — mark it up, add the tags that matter on your line, and keep the signed copy.